DEPI | Digital Egypt Pioneers Initiative | Junior Data Analyst track

Egypt Residential Real Estate Power BI

Real Estate Dataset Analysis: from raw data to an interactive report

Team: Saleh | Maged | Abd Elkhalek | Sama Raed | Sama Abdelmeged | Mostafa

Project overview

Build an interactive Power BI report on Egyptian residential real estate transactions, using a star schema of one fact table (25,000 transactions) and eight dimensions.

Objectives

- Clear view of sales, pricing, cancellations, agent and channel performance
- Validated, well-modeled dataset with documented metric definitions
- Polished, navigable report published to the agreed workspace

In scope

- Data cleaning
- Star-schema modeling
- DAX measures
- Report pages and interactivity
- Validation, publishing, documentation

Out of scope

- Live data feeds
- Predictive models
- External data sources

25,000	8	2022-2025	EGP
transactions (fact rows)	dimension tables	calendar coverage	all amounts in Egyptian pounds

The dataset is synthetic but internally consistent (per its README); it is not an official market extract.

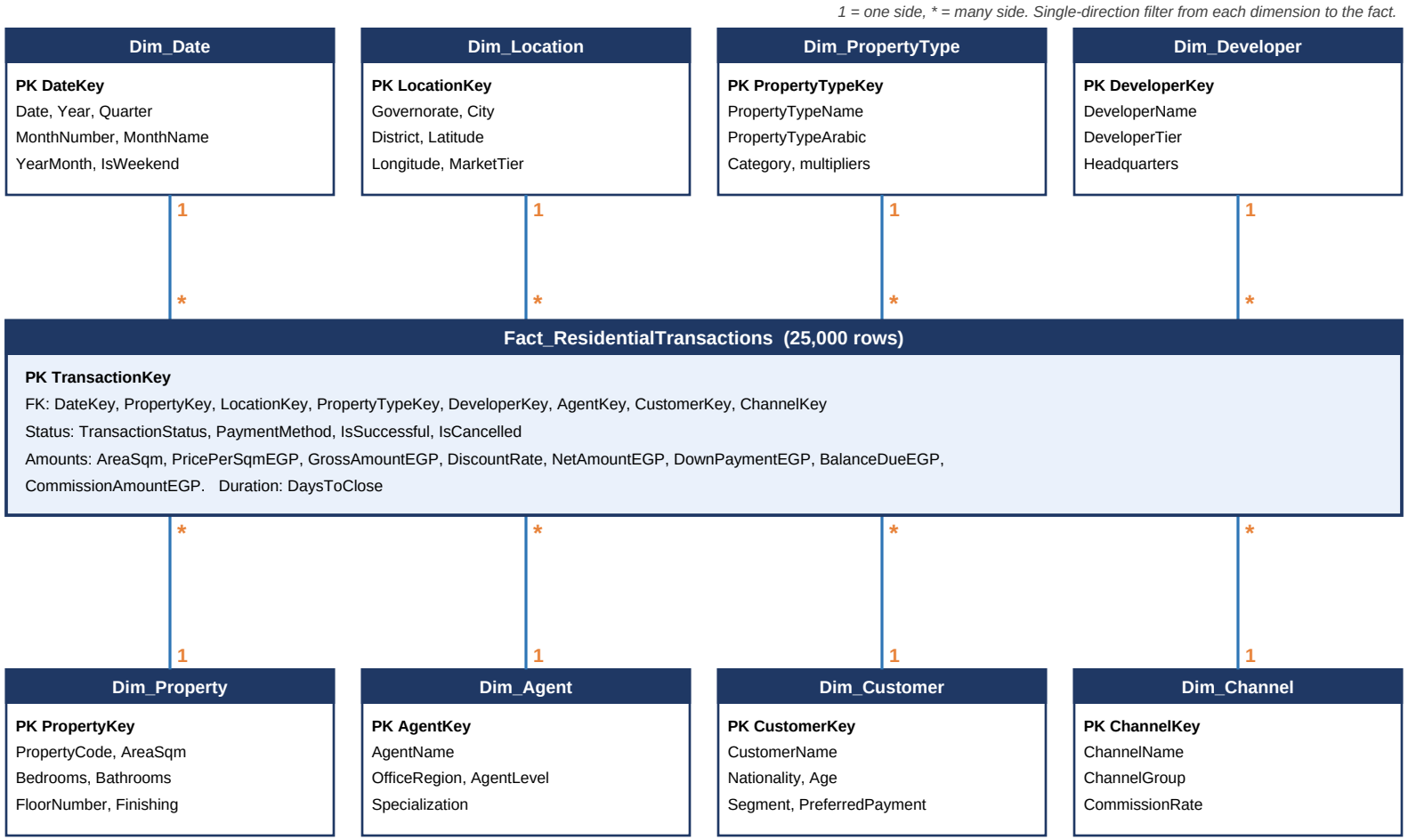
Team and workflow

Six members, clear ownership

Member	Technical responsibility	Role	Key outputs
Saleh	Power Query and data preparation	Project Coordinator	Clean tables, data-quality log, task tracker
Maged	DAX measures and KPIs	Quality Reviewer	Measure library, review feedback
Abd Elkhalek	Report design, pages, KPI cards	Team Leader, Presenter	Report pages, final presentation
Sama Raed	Data modeling, relationships	Requirements Lead	Requirements doc, star-schema model
Sama Abdelmeged	Validation and publishing	Documentation Lead	Validation checklist, published report, handover
Mostafa	Slicers, drill-through, navigation	Meeting Facilitator	Interactive report, minutes

Handoff flow: requirements (Sama Raed) > data (Saleh) > model (Sama Raed) > DAX (Maged) > design and interactivity (Abd Elkhalek, Mostafa) > review (Maged) > validate and publish (Sama Abdelmeged) > present (Abd Elkhalek)

Data model: star schema



Modeling decisions

- One-to-many, single direction from each dimension to the fact
- Fact grain: one transaction or booking, one property, one date
- Dim_Property repeats three keys; relate only the fact to Location, Type and Developer
- Dim_Date marked as the date table (1,461 days)
- Use CustomerKey, never CustomerName (252 names for 2,500 customers)

Data sources and source of truth

Nine CSV files, one workbook copy

File	Rows	Columns	Role
Fact_ResidentialTransactions	25,000	22	Fact table
Dim_Date	1,461	8	Calendar 2022-01-01 to 2025-12-31
Dim_Property	3,000	11	Residential inventory
Dim_Customer	2,500	6	Customers
Dim_Agent	120	5	Agents, region and level
Dim_Location	30	9	Governorate, city, district, tier
Dim_Developer	20	4	Developers and tiers
Dim_PropertyType	7	7	Types with Arabic names
Dim_Channel	5	4	Channels and commission rates

Source of truth: the CSV files. The StarSchema workbook has identical values in all nine sheets (compared column by column). The existing PBIX could not be opened for its Power Query steps, so what it imported is unconfirmed. Original files are never modified.

Data quality audit

Full-table profiling of every column

0	0	0	0	1
nulls, blanks, placeholders	exact duplicate rows	orphan foreign keys	negative or invalid values	corrupted text value

Real defect and open items

- **DQ-01:** City for LocationKey 10 is garbled Arabic/Latin text; 675 fact rows (2.7%) join to it. Repaired to "Hadayek October" after approval
- **DQ-02:** "6th of October" sits under Cairo and Giza
- **DQ-04:** weekend flag uses Sat/Sun, not Fri/Sat
- **DQ-08:** 563 price outliers (2.25%) are plausible; kept

Rules verified in all 25,000 rows

- Gross = area x price per sqm
- Net = gross x (1 - discount); zero when cancelled
- Balance = net - down payment
- Commission only on completed deals
- Fact keys agree with Dim_Property

13 issues logged in total; most are informational notes for modeling and DAX, not data errors.

Data preparation in Power Query

Saleh: import, clean, transform, validate

Approach

- Raw staging queries (stg_) load each CSV unchanged
- Clean queries keep the original table names, so relationships and DAX still bind
- Explicit data types, en-US culture
- Parameters: source folder and a switch for the City fix
- Error guard on the fact table: fails loudly, no silent row loss

What is NOT done

- No rows removed
- No values imputed or set to zero
- No outliers deleted
- No columns added or removed
- Cancelled and Pending zeros kept as real values

Before and after

- Row and column counts identical for all 9 tables
- Exactly 1 cell changed in total (City, LocationKey 10)
- Non-text characters left in Dim_Location text: 0
- Verified with a Python replica of the same logic

Honest status: the M code is generated and not yet run in Power BI Desktop; manual verification in Desktop is the remaining step.

Independent validation

Existing report re-checked

validation_report.csv: 11 of 11 checks recalculated

- Row count, unique keys, key coverage (property, location, date)
- Net never above gross; balance logic; cancelled have zero net
- Completed sales positive; dimension keys unique; Egyptian governorates
- All checks hold in the data

Extra checks added (all pass) and gaps found

- Gross, net, commission formulas and fact-versus-property consistency: 0 mismatches
- Gap: the report does not test text corruption (DQ-01)
- Gap: it does not test city-governorate consistency (DQ-02)
- Updating the team self-check from "all checks passed" to include these items is recommended

Checks passed does not mean nothing left to fix: one defect and five approval decisions remain open.

What the data shows so far

Draft facts, for Maged to confirm in DAX

15,484
completed (61.9%)

7,525
pending or reserved (30.1%)

1,991
cancelled (8.0%)

EGP 52.8 bn
completed net sales

Metric definitions matter

- README "Total Net Sales" includes Pending and Reserved: EGP 79.3 bn, 33.4% of it not yet completed
- Average price per sqm: all rows give EGP 18,395; completed only gives EGP 19,755 (6.9% higher)
- Label cards "Completed net sales"

Handle with care

- Cancelled rows keep gross value and area
- DaysToClose is 0 for Pending and Cancelled: exclude from averages
- Price per sqm has legitimate high-end outliers (villas, premium areas)
- PreferredPaymentMethod is not the transaction method

Requirements

Led by Sama Raed

9	12	15	12
stakeholder groups	user stories, 7 use cases	functional requirements	non-functional requirements

Who needs what

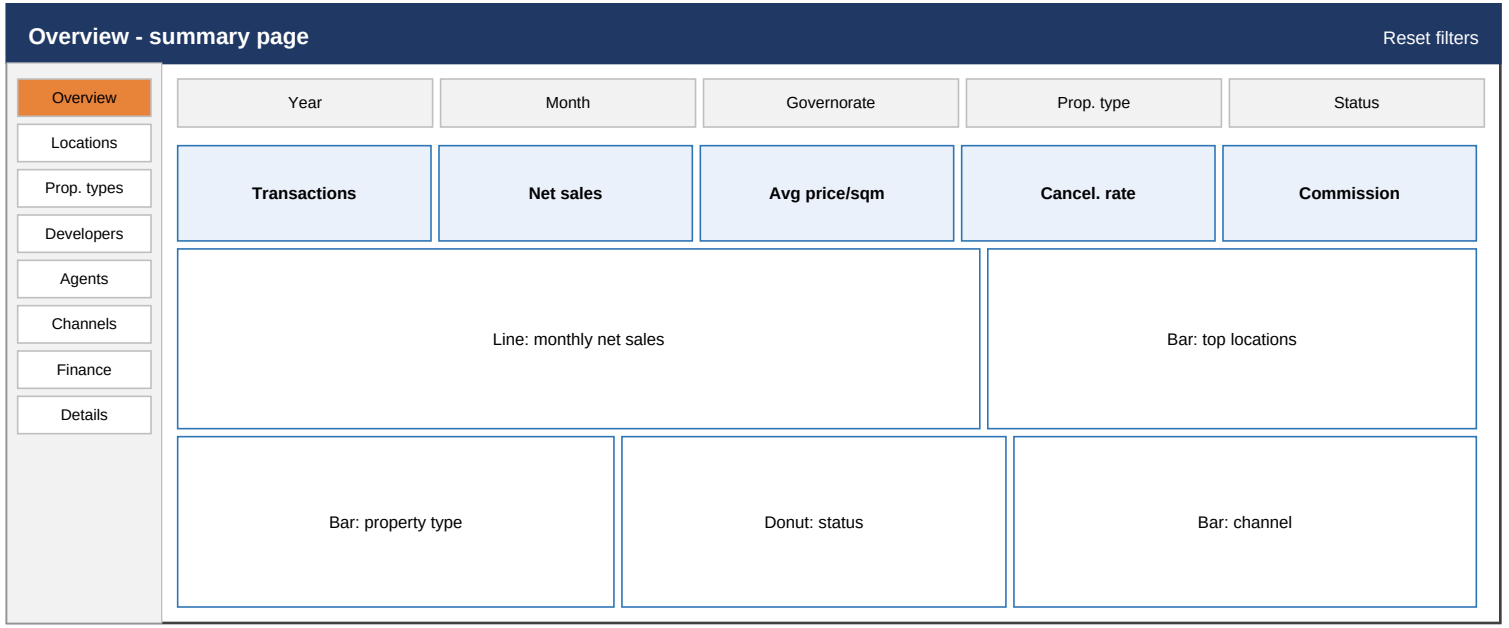
- Sales managers: overall performance, markets, cancellations
- Team leaders and agents: rankings, commission, days to close
- Marketing, pricing, developer relations, finance
- Lecturer: documentation, correct metrics, clear presentation

Key non-functional targets

- Pages respond in under 5 seconds (25,000 rows)
- Measures match source totals 100%
- Report availability of at least 99%
- Accessible design: contrast, alt text, tab order

Report design: pages

Wireframe: Overview page (low fidelity)



Eight pages

- Overview, Locations, Property types, Developers
- Agents, Channels, Finance
- Details: drill-through to transactions
- Same grid on every page: slicers on top, KPI cards, charts, navigation at the left
- Reset-filters and back buttons in fixed positions

UI/UX guidelines

Proposed, for design team approval

Navy	Blue	Teal	Orange	Green	Amber	Red
11.6:1 on white	4.8:1 on white	5.2:1 on white	7.7:1 dark text	5.0:1 on white	5.5:1 on white	5.4:1 on white
Structure, headers	Main data series	Second series	Highlights (dark text)	Completed	Pending	Cancelled

Layout and type

- One grid, same navigation everywhere
- Segoe UI; titles 16-18 pt, card values 24-28 pt, text at least 10 pt
- About 8 chart visuals per page at most

Interactivity

- Synced Year and Month slicers
- Drill-through to Details
- Bookmarks, tooltips, reset-filters button

Accessibility

- Contrast ratios calculated (4.5:1 minimum)
- Never colour alone: labels and legends
- Alt text, tab order, mobile layout

Plan, milestones and deadlines

Deliverable (official guideline)	DEPI deadline
Project Planning & Management	16 Oct 2026
Literature Review	16 Oct 2026
Requirements Gathering	16 Oct 2026
System Analysis & Design	6 Nov 2026
Implementation (Source Code & Execution)	30 Nov 2026
Final Presentation, Testing & Reports	4 Dec 2026

All documents are to be uploaded to GitHub, per the official guideline.

Team milestones

- M1: requirements, data loaded
- M2: clean data ready
- M3: relationships validated
- M4: core measures
- M5: full report draft
- M6: validated, published, presented

Risks and success KPIs

Each with a named owner

Risk	Impact	Owner
Data quality issues	High	Saleh
Broken relationships	High	Sama Raed
Incorrect DAX (cancelled vs completed)	High	Maged
Missed deadlines, unclear ownership	High	Saleh
Publishing or access problems	High	Sama Abdelmeged
Cluttered design; broken interactions	Medium	Abd Elkhalek, Mostafa

Project success KPIs

- Milestones on time: 90% or more
- Null or invalid values in key columns: 0
- Relationships validated: 100%
- Measure accuracy vs source: 100%
- Page response time: under 5 seconds
- Interaction tests passed: 100%
- Stakeholder satisfaction: 4 of 5 or more

Status and next steps

Done

- Planning, literature review and requirements documents
- Data audit, validation recheck, 13-item issue log
- Cleaned copies and Power Query M code
- Database and UI/UX design pages

In progress

- Applying the M code in Power BI Desktop (not yet verified there)
- Data model and relationships
- DAX measures and report build

Decisions needed

- DQ-01 city label
- DQ-02 governorate for Location 6
- DQ-04 weekend definition
- DQ-07 average price per sqm definition
- DQ-08 keep price outliers
- Remaining design diagrams due 6 Nov